Regulation and Safety

Station BlackOut - Safety Analysis

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June 18, 2026

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Introduction

The aim of this report is to present the first part of the Station Black-Out (SBO) project developed with RELAP5 as a system code. The main purpose of this first delivery is to study the reference model before any new safety-system design is introduced. In particular, the work is divided in two tasks: first, the verification of the steady-state conditions of the plant model and second the analysis of the SBO base case transient. The steady-state analysis is used to check that the model starts from physically consistent and stable operating conditions. The main thermal-hydraulic parameters are compared with the target values provided in the project statement, and their time evolution is checked in order to confirm that the initial condition is sufficiently stable for the transient calculations. After the steady-state verification, the SBO base case is analyzed. This case represents the reference accident scenario and is used as the starting point for the following parts of the project. The objective is to understand the sequence of events, the system response and the main thermal-hydraulic trends during the transient.

Methodology

The reference plant selected for this analysis is the Zion Nuclear Power Station (United States), which was decommissioned in 1998. Although the real plant originally had four primary loops, the RELAP model used in this project is simplified into two equivalent loops only. One loop represents the broken loop and keeps the dimensions of a single loop, while the other represents the intact side and groups the remaining three loops into one single equivalent loop.
Figure 1 shows the nodalization used in order to represent in the best way possible the Zion nuclear power plant.

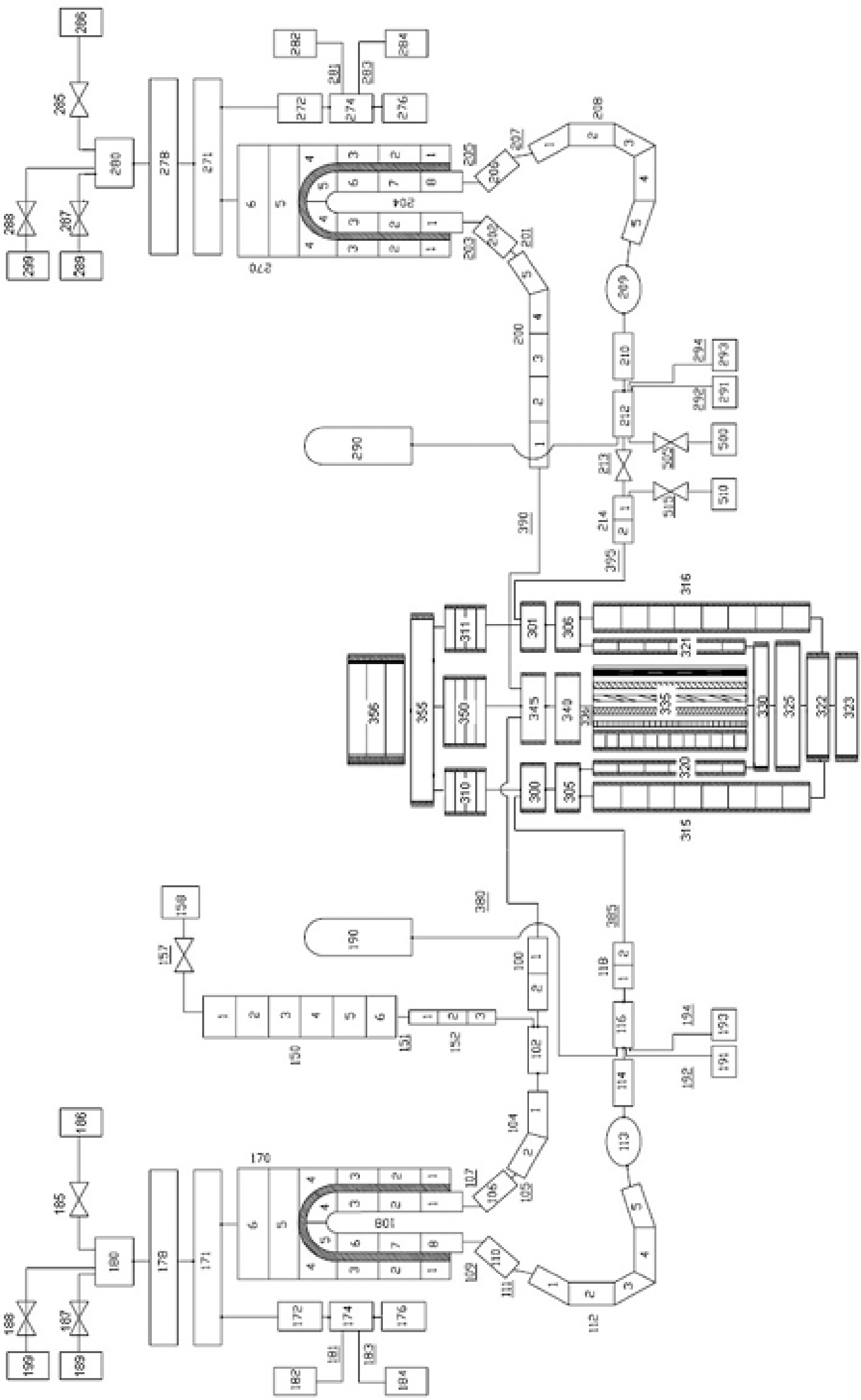

Figure 1: RELAP5 nodalization of the Zion Nuclear Power Plant model.

1 Steady State Analysis

Before starting the accident analysis, it is necessary to verify that the model is in a stable and realistic initial condition. A correct steady state is important because the transient results strongly depend on the quality of the initial condition. The steady-state run is checked by comparing the calculated values with the reference values and by observing the time evolution of the main parameters. If the variables remain approximately constant and the deviations from the target values are acceptable, the initial condition can be considered suitable for the transient simulation.

Parameter	Expected value	Steady State Results
Power (MW)	3250	3250.0
Primary pressure in the hot leg (MPa)	15.5	15.5
Secondary pressure (MPa)	6.7	6.71
Pressurizer level (m)	8.8	8.86
Broken Steam Generators DC level (m)	12.2	12.2
Intact Steam Generators DC level (m)	12.2	12.2
Core outlet temperature (K)	603	603.4
Core inlet temperature (K)	571	570.8
Broken Main feedwater flow (kg/s)	459	455.8
Intact Main feedwater flow (kg/s)	1325	1324.4
Intact loop flow (kg/s)	12865	12436
Broken loop flow (kg/s)	4351	4356

Table I: Expected vs Simulation results for Steady State Values

Table I shows the main variables that must be checked in the steady-state calculation. The steady-state calculation shows that the main plant variables are close to the reference values; small deviations may appear because of nodalization choices, control-system settings, and numerical approximations. However, the overall agreement is sufficient to accept the initial condition for the SBO transient analysis.

1.1 Steady State Verification

In this section some plots regarding the steady state case are provided. The purpose is only to ensure that the calculations done with RELAP are coherent with the steady state conditions the plant should be operating at.

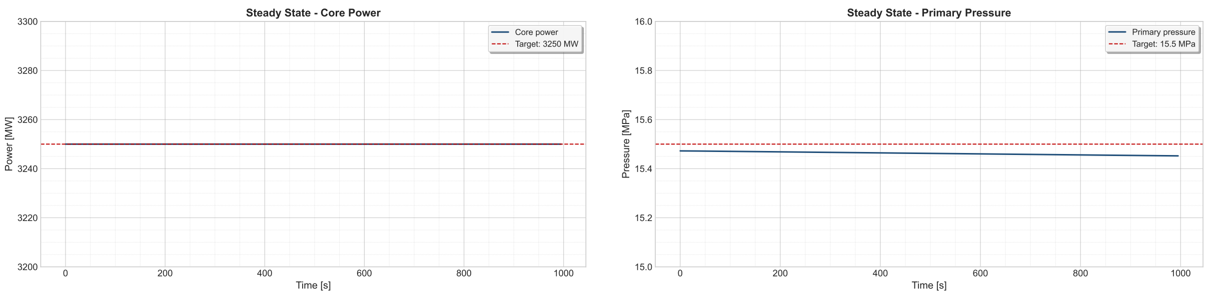

Figure 2: Core Power (on the left) and primary pressure (on the right).

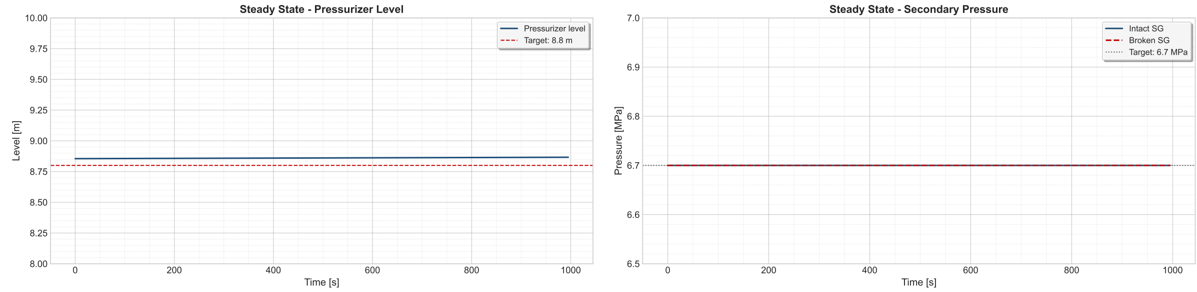

Figure 3: Pressurizer Level (on the left) and Secondary Pressure (on the right).

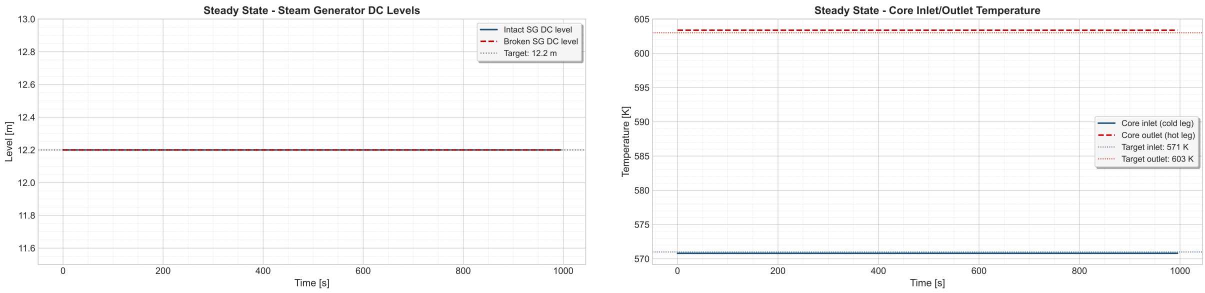

Figure 4: SG levels (on the left) and RCS temperatures (on the right).

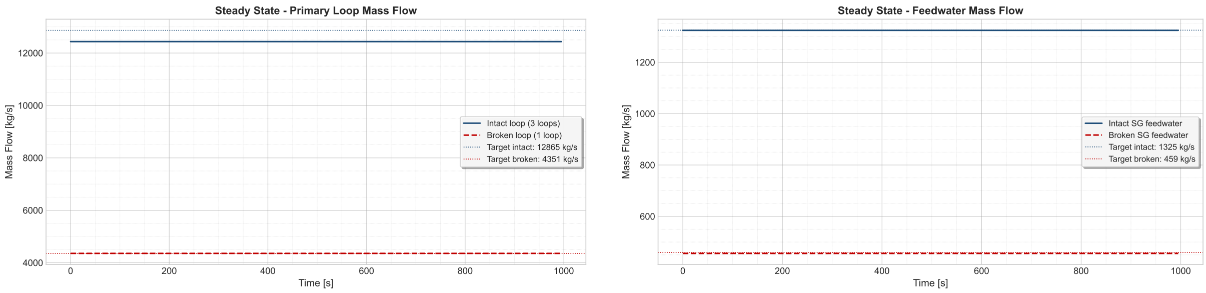

Figure 5: Primary Mass Flow (on the left) and FW Mass Flow (on the right).

As we can see, all the variables analyzed are stable and do not change over the time. Some of them present little uncertainties that are probably due to numerical errors and the complexity of the phenomena involved.

2 Base Case Analysis

After the steady-state verification, the SBO base case is studied. The Station Black-Out is assumed to begin at $t = 0$ s with aggravated conditions. Under these conditions, every system that depends on electrical power is considered unavailable. As a result, the reactor coolant pumps trip, the control rods are inserted, and other relevant functions, such as the cooling of the pump seals, are lost. The general assumptions that are common to all the considered accident sequences are summarized in Table II below.

Table II: Main events to be included in the SBO base case description.

Event / action	Short description
SBO initiation	Loss of off-site and on-site AC power according to project assumptions
Reactor scram	Automatic shutdown following the transient logic
RCP trip	Reactor coolant pumps stop after loss of power
FW / AFW behavior	Feedwater and auxiliary feedwater response according to the base-case logic
Pressurizer systems	Heater and control-system behavior during SBO
Safety injection unavailability	HPSI and LPSI unavailable during the transient
Steam line / relief actions	Valve actions according to the project assumptions
Additional operator actions	To be completed based on the adopted base-case procedure

In order to simulate all the accidental sequence, some actions that could be performed from the operators have been taken into account. These are listed below.

Action / Event	Description
RCP Seal Leak	Starts 10 minutes after SBO initiation with a leak size of 0.00001 m2 per pump.
Leak area increase	The leak area increases 4x when saturation takes place at the RCP.
TDP AFW Injection	Auxiliary feed water is provided by a turbo-driven pump (TDP) which is controlled manually and requires batteries.
TDP battery depletion	Batteries last for 5h, after which the TDP becomes uncontrolled, injecting water until the separator is flooded and the TDP stops.
Cooldown strategy (EOP)	Operators follow a strategy to cool down the SG at a rate of 55 K/h until 15 bar.
Primary depressurization (EOP)	Operators fully open the Pressurizer relief valves to depressurize the primary system if the Core Exit Temperature is higher than 923 K.
AC power recovery	Recovery of AC power occurs after 12h, which recovers AFW, HPIS, and LPIS.

Table III: Configurable actions and conditions for the SBO sequences in the Zion model

2.1 First Analysis

The first analysis was performed considering that all the EOPs listed in Table III have taken place. We are going to analyze the transient in order to verify the situation of the plant after the SBO has occurred.

2.1.1 Reactor Coolant System

Right after the SBO event, the SCRAM immediately shuts down the reactor by inserting the control rods, leading to a rapid decrease in power, that is plotted in Figure 6. The residual value present in the graph is due to the decay heat, which represents the main issue in this types of scenarios. Moreover, the loss of AC power causes the immediate trip of the pumps.

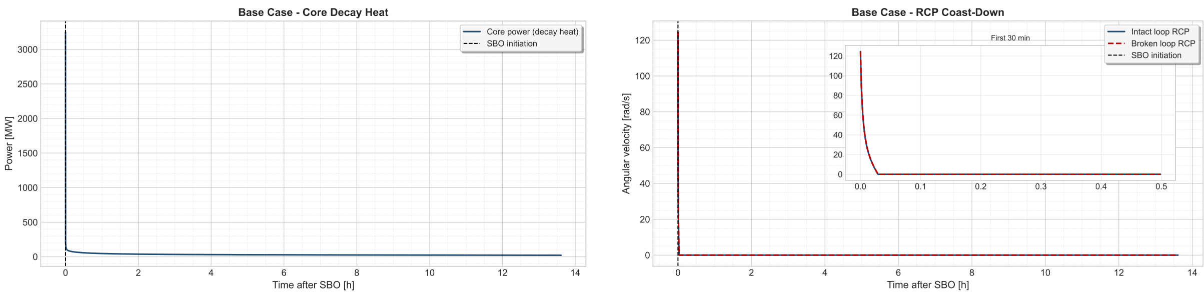

Figure 6: Core Power (on the left) and RCP velocities (on the right).

The sudden power decrease causes a contraction within the coolant, which lowers the level inside the pressurizer, as it is shown in Figure 7. The level reaches zero and it starts to increase again after 12 hours, when the AC power is recovered and water is introduced again by the means of HPIS, LPIS and the pumps. The loss of level is also enhanced by the loss of inventory, as the seal injection system is not available during a SBO. This lead to some leaks from the pumps, which contributes to the decrease. However, an empty pressurizer does not imply that the core is uncovered. This situation is properly described in Figure 7.

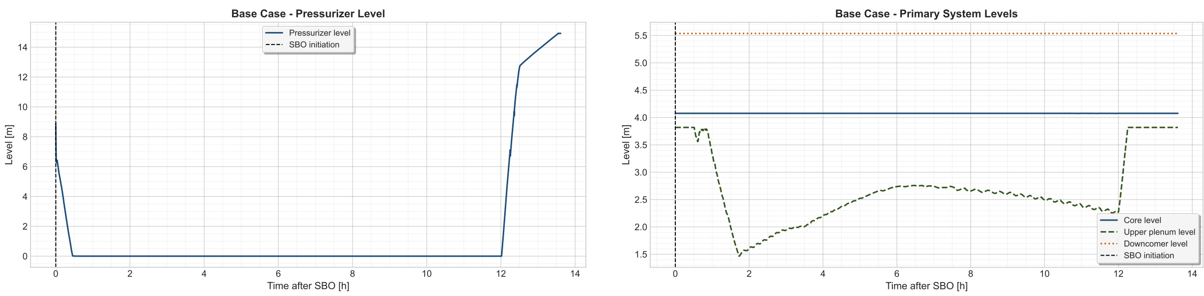

Figure 7: Pressurizer Level (on the left) and Core Levels (on the right).

In Figure 8 the pumps's leaks is shown.

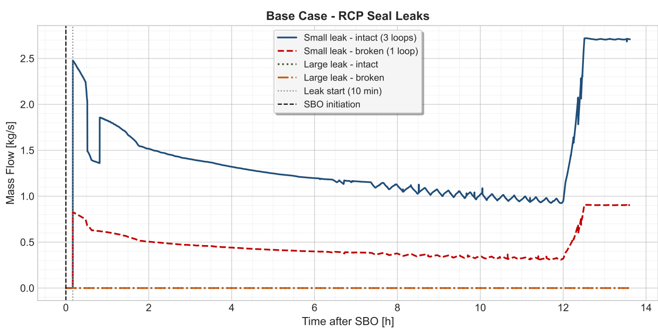

Figure 8: RCP Seals Leaks. Notice that the intact loop represents 3 loops.

According to the EOPs, if the outlet temperature from the core is higher than 923 K, the operators should depressurize the primary system using the PRZ valves. In our case, this procedure has not been necessary, as the core exit temperature does not overcome 923 K. Figure 9 shows the main parameters related to the temperatures of the core.

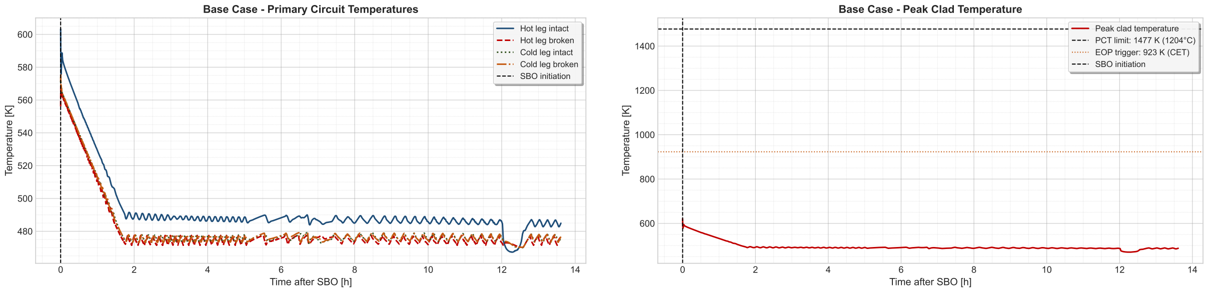

Figure 9: Core Temperatures (on the left) and Cladding Temperature (on the right).

The primary temperatures are lower than the threshold limit established for the operator, as well as the peak cladding temperature. It remains lower than the safety limit, preventing damages to the cladding. It also indicates that DNBR is prevented and the core is covered. The constant behaviour can be explained as a consequence of some natural circulation phenomena that could have taken place inside the primary, due to the pressure difference between primary and secondary. Figure 10 shows that a little flow inside the RCS is present, even if the pumps are tripped. The natural circulation guarantees the heat exchange between the core and the SGs.

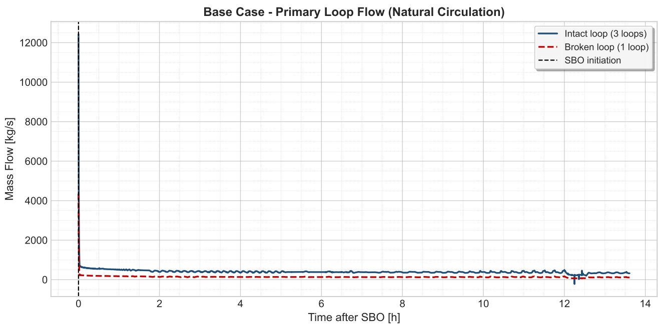

Figure 10: Primary Mass Flow

2.2 Secondary System

It is finally the moment to take a look at the behaviour of the secondary system. In a SBO scenario, this system plays a fundamental role, since it is used to extract the heat coming from the core where no AC powered system is available. A natural circulation flow regime is established by the pressure's difference between the primary and the secondary, helped by the elevation of the latter. In fact, the SG is usually located in a higher position than the core, in order to stimulate the gravity forces that act by density.

As we said in the previous chapter, a cooldown procedure is performed by the operators. The aim is to cool down the SG at a rate of 55 K/h, until a pressure of 15 bar (1.5 MPa) within the secondary is guaranteed. In Figure 11 this situation is well described.

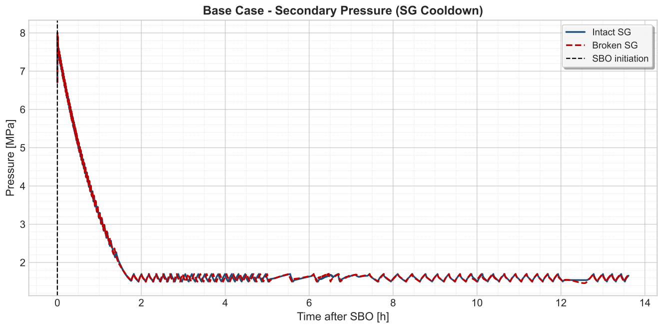

Figure 11: SG Pressure Evolution during the Transient.

The procedure is done by the relief valves, located at the top of the SG. Their function is to release steam to the atmosphere, in order to reduce the pressure inside the SG. As we can see from the picture, the procedure is well carried out until the target pressure of 15 bar is reached.

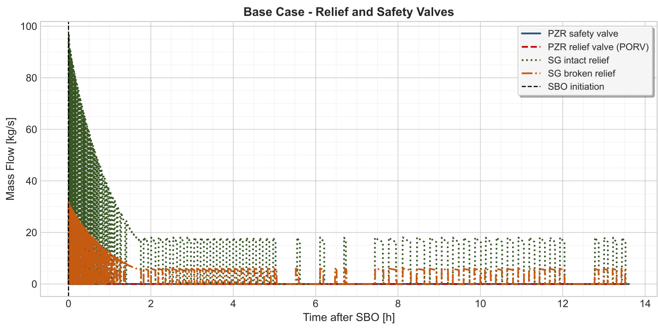

Figure 12: Relief Valves Flow. Notice that the flow coming from the PRZ valves is zero, as the primary depressurization has not been carried.

Figure 12 shows the flows coming from the relief valves. The behaviour of the valves is coherent with the SG pressure curve, the most of the steam is released at the beginning of the procedure and gradually decreases. After that, the flow remains constant, since the operators aimed to keep the SG pressure constant and the Auxiliary FeedWater is constantly injecting water. The latter is shown in Figure 13. The batteries that power the turbo-driven AFW last for 5 hours. As mentioned in Table III, at this moment the TDP becomes uncontrolled and injects water until the SG separator is flooded.

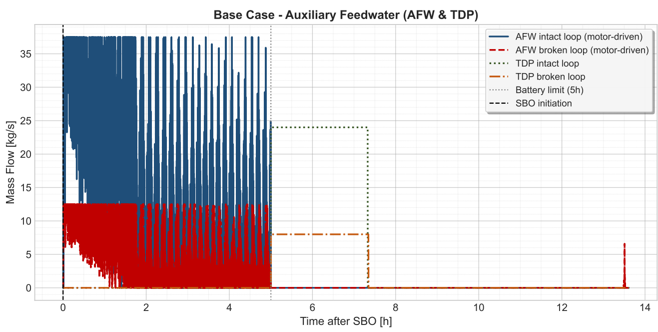

Figure 13: Auxiliary Turbo-Driven Feedwater Flow. Remember that the intact loop represents 3 loops.

3 Additional Calculations

As part of the base case simulation, there are two operator procedures assumed to be preformed. These are the steady depressurization of the secondary system, and the depressurization of the primary based on a high outlet core temperature. Apart from the complete base case, the simulation was also performed without these operator interventions in order to gauge their effectiveness. The second EOP mentioned above, pertaining to primary depressurization, was never actuated in the base case as the outlet temperature never reached the threshold value. Therefore, this case was identical to the base case and will not be analyzed further and only the additional case without the EOP 1 is analyzed below. This simulation analyzes the dynamics and critical values of the system when the steam generators are not depressurized allowing for a further reduction in primary temperature and pressure.
Figure 14 below demonstrates the lack of pressure relief of the secondary system. The lack of the depressurization EOP maintains the pressure high in the secondary system. This graph verifies that the EOP is not initiated as intended. In the base case, the relief valve setpoint is progressively reduced from the nominal value of 6.7 MPa down to 1.5 MPa, following the 55 K/h cooldown ramp. Without this operator action, the setpoint remains fixed at the nominal opening value of 7.58 MPa throughout the entire transient. As a consequence, the secondary pressure does not decrease monotonically but instead exhibits a strong cyclic behaviour: as the decay heat continuously produces steam, the pressure rises until the relief valve opens at 7.58 MPa, releasing steam to the atmosphere. Once the pressure drops below the closing setpoint of 7.79 MPa, the valve closes and the pressure begins to rise again. This open-close cycling of the relief valves repeats continuously, maintaining the secondary pressure oscillating between approximately 4.5 and 8.0 MPa rather than reaching the 1.5 MPa target of the EOP.

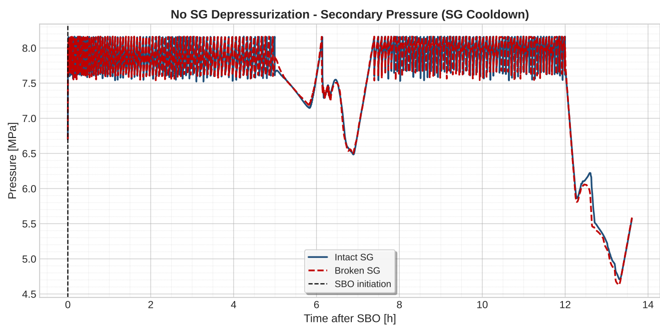

Figure 14: Secondary Pressure.

One of the main concerns during the transient event is the protection of the fuel cladding. Figure 15 shows the temperature of the cladding as a way to verify a lack of oxidation of the zirconium. The temperature in the primary does not reach excessive levels. Although it is higher than the base case which sees a drop in temperature of the cladding shortly after the event initiation, the temperature is still a safe margin away from rapid oxidation.

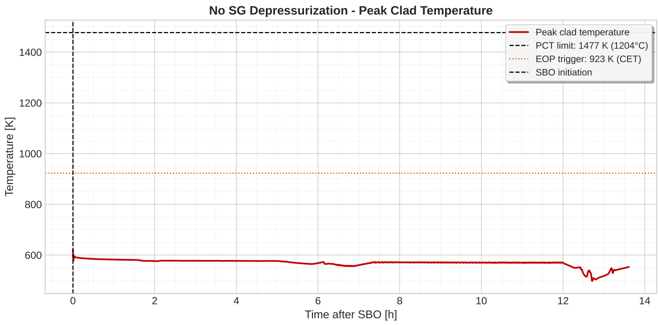

Figure 15: Peak Cladding Temperature.

Another factor of concern is the pressure in the primary. Without the further cooldown of primary system as a result of the steam generator depressurization, the pressure in the primary is much more elevated. One of the main concerns of this value is the correlation between leak rate and pressure. With higher primary pressures, we can expect to lose more inventory through the RCP seals. This situation is well represented in Figure 16.

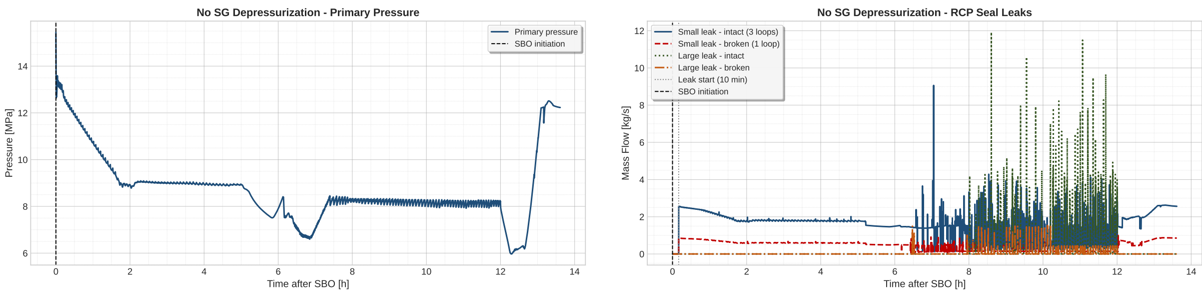

Figure 16: Primary pressure (on the left) and RCP seal leak rate (on the right).

As discussed above, the leak rate of the RCP seals is of concern in this event, and becomes more of an issue with higher primary pressures. Figure 16 confirms this. The higher primary pressure has a direct consequence on the RCP seal leak rates. Since the leak flow is driven by the pressure difference between the primary circuit and the containment atmosphere, a higher primary pressure results in a significantly larger leak flow. The seal leak rates in the no-cooldown case reach peak values of approximately 8-10 kg/s for the intact loop, compared to roughly 2.5 kg/s in the base case — about 3-4 higher. The irregular oscillating pattern of the leak rate directly correlates with the cyclic pressure oscillations of the secondary system described above. Despite this increased inventory loss, no core uncovery is observed within the 12-hour simulation window, indicating that the plant remains in a safe condition. However, in a longer SBO scenario, the sustained high leak rate would represent a more serious concern for primary inventory management.
The simulation demonstrated that the EOP does help to reduce the temperature of the core and the pressure of the primary system. However, based on this analysis is not essential in order to protect the integrity of the fuel cladding during a 12 hour station black out.

4 Event tree for the SBO scenario

The event Tree analysis is an inductive procedure that aims to show all the possible outcomes of an initiating event, in our case an SBO with aggravated conditions. The objective is to identify all the potential accident scenario in a complex frequency. At the end of the process, it is possible to calculate the core damage frequency by assigning a frequency to each sequence. In this part an event tree for the SBO scenario regarding the ZION nuclear power plant is presented, together with the PSA frequency calculation.

4.1 Initiating Event and Top Events considered in this analysis

The initiating event is the SBO occured in aggravated conditions. It implies that all the systems that require power are unavailable. The top events that were chosen to be analyzed in this event tree are listed in Table IV.

Table IV: Top events considered in the SBO event tree.

ID	Top event	Description	Timing	Success criterion
TE1	Small leak	RCP seal LOCA induced by loss of seal injection and thermal barrier cooling. Break area 1×10^{-5} m^2 per pump (intact loop counts as three pumps).	$t = 10$ min	Seal integrity maintained; no leak develops.
TE2	AFW & TDP	Turbine-driven pump delivers auxiliary feedwater to the steam generators, providing the only available secondary heat sink during SBO.	$t = 0^+$ (on demand)	TDP starts and supplies sufficient feedwater to both SGs.
TE3	Large leak	Amplification of the seal leak by a factor of four when saturation conditions are reached at the RCP location, according to NUREG/CR 7110 Vol. 2.	$t = t_{\text{sat,RCP}}$	Leak area does not amplify (or saturation not reached at the seals).
TE4	Extra batteries	DC power supply enabling manual control of the TDP and key instrumentation. Beyond depletion the TDP runs uncontrolled until the SG separator floods and trips it.	$t = 5$ h	Batteries provide controlled TDP operation until AC restoration.
TE5	AC restoration	Recovery of off-site or emergency on-site AC power, restoring AFW, HPSI, LPSI and component cooling.	$t = 12$ h	AC power restored before core uncovery.

Chosing 5 top events for the event tree would yield 32 possible sequences. However, some sequences can be discarded as their physical meaning stop being sensible in case of failure:

- **Loss of TDP collapses all downstream branches**. In case Turbo-Driven Pumps are not available right after the initiating event has taken place, a quick dry-out of the steam generator may occur. This would lead to lose the major heat sink to the secondary, ending up in core uncovery before any following event;
- **Large leak is logically disjoint from no small leak**. TE3 represents the 4 amplification of the existing small seal leak when saturation is reached at the RCP. If TE1 = success (no seal leak ever develops), there is no leak to amplify. The two events are not independent. The Large Leak top event is therefore not branched in the upper half of the tree, and the path goes directly from AFW & TDP success to Extra Batteries. This eliminates 4 phantom sequences that have no physical meaning.

4.2 Expected Results

This section presents the results we expect based on the expected physical behavior of the plant. Each sequence is derived taking into account the success criteria of each TE and the most likely outcome in case of failure or success. They can be seen in Figure 18. The lower half of the picture represents the possible outcomes in case of small leak (TE1) occurs, while the upper part represents the other possible path. As we said before, is TE1 = success, which implies no small leak occurring, a large leak is not going to amplify an already existing break (cause there is none) and therefore the two events are not indipendent. This "disjoint" is indicated at the corresponding place within the graph.
In case extra batteries fail, together with no recover of the AC power, we expect the system to have core damage, as it would not be possible to cool the core. Moreover, if extra batteries are available but no AC power is recovered, core damage is expected to take place as the batteries would get depleted while operating. The unavailability of the Auxiliary Feedwater System and Turbo-Driven pumps would lead to core damage as the main heat sink present within the plant - the steam generators - would be lost.

At the lower part, the sequences are extrapolated following the same scheme applied on the upper part. Core damage will occur if no AFW & TDP are available. Depending on the availability of the extra batteries and the capability of recovering AC power, the predicted outcomes are drawn.
The RELAP analysis will be used either to confirm this configuration or assess different types of events and outcomes.

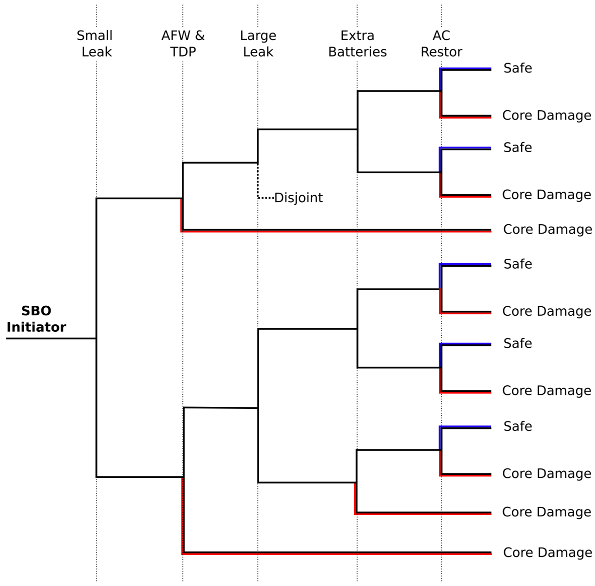

Figure 17: Expected Event Tree.

4.3 Results

Table V: Results of the simulated sequences: Y/N indicates success/failure. Core Damage is obtained when the peak cladding temperatures is higher than 1477K (1204 °C).

Seq.	**TE1** SL	**TE2** AFW	**TE3** LL	**TE4** Batt	**TE5** AC	**Outcome**	**PCT [K]**	t_{CD} **[h]**
1	Y	Y	Y	Y	Y	SAFE	621	—
2	Y	Y	Y	Y	N	CD	1702	22.52
3	Y	Y	Y	N	Y	SAFE	621	—
4	Y	Y	Y	N	N	CD	1701	18.35
5	Y	N	Y	Y	Y	SAFE†	963	—
6	N	Y	Y	Y	Y	SAFE	621	—
7	N	Y	Y	Y	N	CD	1701	21.19
8	N	Y	Y	N	Y	SAFE	621	—
9	N	Y	Y	N	N	CD	1701	17.39
10	N	Y	N	Y	Y	SAFE	621	—
11	N	Y	N	Y	N	CD	1701	21.32
12	N	Y	N	N	Y	SAFE	621	—
13	N	Y	N	N	N	CD	1701	17.42
14	N	N	N	Y	Y	CD	1478	12.21
15	N	N	N	Y	N	CD	1701	12.29

Table V presents the RELAP simulation's results. Several considerations can be made regarding the difference between expected and actual results:

- AC and AFW recovery is the dominant factor that determines whether core damage will occur or not. In all the scenarios when AC and AFW are available, the PCT does not overcome 621 K. The combination of AFW and AC recovery is sufficient to prevent core damage. Once AC power is restored, HPIS and LPIS can keep on cooling the core.

- Sequence 5 (ET5) shows that, even though the AFW is not available, in a condition without leakages through the pump's seal the batteries are able to avoid core damage until the AC power is recovered. This can be explained taking a look at the PCT: without AFW, the temperature rises until the PORV valve starts operating, releasing steam from the pressurizer in order to depressurize the primary. This is different from what we have considered in the preliminary event tree, where the unavailability of AFW determines core damage indipendently from the other systems availability.

- The amplification of the leak, that takes place when the water reaches saturation temperature, has a small effect on the outcome of that sequence. For instance, if we compare ET07 to ET11 or ET09 with ET13, the difference in time is relatively small. This behavior suggest that the main mechanism that lead to core damage is the heat balance through the steam generators, while the leak from the RCP plays a marginal role in the behavior of the plant.

- ET14 can be considered as a "bordeline" case, since its PCT reaches 1478K, slightly above the safety limit. This demonstrates that in presence of leaks from the seals and no AFW, the batteries are not sufficient to prevent core damage. The time needed to reach PCT is 21 minutes higher than the recovery of the AC, which occurs at 12h after the SBO event.

In Figure 18 all the sequences are plotted, as a function of the peak cladding temperature and the time to reach core damage.

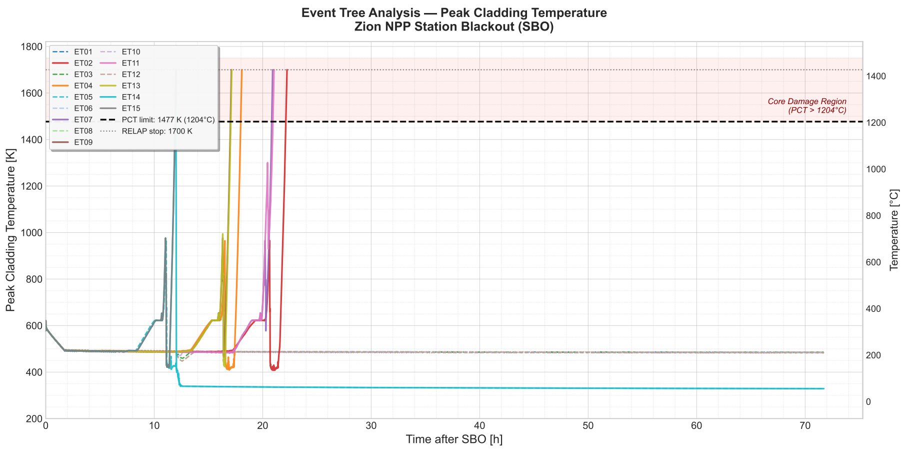

Figure 18: Obtained Event Tree.

5 Probabilistic Safety Analysis

The creation and verification of the event tree for the SBO case allows for the calculation of the contributions of the SBO to the Core Damage Frequency and to identify the Conditional Core Damage Frequency (CCFD).

5.1 Mathematical Representation

The first step to analysis the event tree qualitativly is identify the initating event frequency and the conditional probabilities of each of the elements of the event tree. The key variables under consideration are as follows:

I	Initiating Event (SBO)
LT	Long-term SBO
SB	Short-term SBO
A	AFW & TDP
S	Small Leak
L	Large Leak
E	Extra Batteries
R	Restoration of AC

In order to calculate the probability of each individual event, first the probability of failure of a single component is calculated. This calculation is followed by a calculation to determine the probability of at least 3 loops working. This is because we consider a conservative approach in our modeling and consider only 3 of our system working. The equation to transform the probability of an individual component failure ($P(F_C)$) to the overall system operability ($P(O)$) is as follows:

$$P(O) = \left(\overline{P(F_C)}\right)^4 + \left(\overline{P(F_C)}\right)^3 P(F_C) \binom{4}{1} \tag{1}$$

Initiating Event

The initiating event frequency of the SBO is derived from the NUREG CR-7110 Vol 2 [1]. From this report the frequency of Long-Term SBO frequency is identified as 2x10^{-5} pry (per reactor year), and the Short-Term SBO frequency is defined as 2x10^{-6} pry. One interesting finding from this is the higher frequency of long-term SBOs than short term ones. From the frequencies above the initiaing event frequency is derived below:

$$\begin{aligned} F(I) &= F(LT \cup ST) \\ &= 2 \cdot 10^{-5} + 2 \cdot 10^{-6} \\ &= 2.2 \cdot 10^{-5} \text{ pry} \end{aligned} \tag{2}$$

AFW & TDP

The AFW and TDP failure frequency is assumed to be largely driven by the failure of the TDP based on the findings of the ORNL CONF-920375-6 analysis [2]. As such, the probability of failure of this system is taken from the frequency of failure of the TDP system. This system has two failure modes: failure on demand and failure on run. To use a conservative estimate, the sum of the probability of the failure on demand and failure during a mission time of 5 hours is used. The failure on demand probability is:

$$P(A_d) = 4.6 \cdot 10^{-3}$$

The failure on run probality is calculated as follows:

$$\begin{aligned} P(A_r) &\approx \lambda_f \cdot t_m \\ &\approx 8.45 \cdot 10^{-5}\text{hr}^{-1} \cdot 5\text{hr} \\ &\approx 4.225 \cdot 10^{-5} \end{aligned} \tag{3}$$

Giving a total probability of:

$$\begin{aligned} P(A_f) &= P(A_d \cup A_r) \\ P(A_f) &= 4.54\text{x}10^{-5} \end{aligned} \tag{4}$$

Using equation 1, the probability of the operability of the AFW & TDP is:

$$\begin{aligned} P(A) &= \left(\overline{P(A_f)}\right)^4 + \left(\overline{P(A_f)}\right)^3 P(A_f) \binom{4}{1} \\ &= (1 - 4.54\text{x}10^{-5})^4 + (1 - 4.54\text{x}10^{-5})^3 (4.54\text{x}10^{-5})(4) \\ &= 0.99999998 \end{aligned} \tag{5}$$

The data for the failure of the TDP was taken from the IHL TDP Performance Analysis [3].

Small Leak

The small leak event is assumed to happen in the case that the Westinghouse Shut Down Seal fails. A preliminary safety analysis of this component from Westinghouse gives it a conservative failure frequency of 1 in 100 cases giving us the follow probability:

$$P(S_f) = 0.01$$

This leads to an operability probability of:

$$P(S) = 0.99940$$

Large Leak

Large leaks in the RCP seals occur in the situation in which saturation temperatures are reached. The conditional probability of this event during an SBO is listed in the NUREG CR-7110 Vol. 2 report [1], it gives a the following conditional proability.

$$P(L_f) = 0.2$$

For the overall operability of the large leak condition, it is assumed that the system fails even if there only a single large leak in a pump. This gives the following proability:

$$\begin{aligned} P(L) &= \left(1 - P(L_f)\right)^4 \\ &= 0.4096 \end{aligned} \tag{6}$$

The conditional probability is used in this case as the probability of a large leak dramatically increases during an SBO event.

Extra Batteries

This section was the most difficult to model. This is because the existence of additional batteries is either an existing plant aspect or the result of proper load shedding which requires extensive human error analysis. In the end, the probability of the extra batteries has a long impact on the overall CCDP as seen in Equation /refeq:CCDP below. An alternative method to determine this probability is considering the proper implementation of FLEX-DG (portable diesel generators) that are now implemented in all US NPP and the vast majority of the global fleet.
The analysis of the proper operator installation and equipment performance is done in Universidad Polítíecnica de Madrid [4]. This gives the follow probability below where $P(E_o)$ and $P(E_f)$ are the probabilities of operator human error and the probability of equipment functionality respectively:

$$
\begin{aligned}
P(E) &= \overline{P(E_o)} + \overline{P(E_f)} \\
&= 0.015 + 0.2905 \\
&= 0.3055
\end{aligned} \tag{7}
$$

AC Power Restoration

The AC power restoration after 12 hours is estimating using the conditional probability of a long term SBO given an SBO event. This method allows us to obtain the probability that AC power is restored within a 12-hour window. The calculation is done using Bayes Theorem for conditional probabilities:

$$
\begin{aligned}
P(R_f) &= P(LT|I) \\
&= \frac{P(I|LT) * P(LT)}{P(I)} \\
&= \frac{1(2\text{x}10^{-5})}{2.2\text{x}10^{-5}} \\
&= 0.909
\end{aligned} \tag{8}
$$

Similarly to the large leak condition, the Restore AC probability is not subject to equation 1. This is because the Bayesian analysis above is down for at plant level and not at an individual component level. So the probability of success is:

$$P(R) = 0.091$$

5.2 CCDP

The conditional core damage frequency is the probability of core damage given an initiating event happens. This allows for the analysis of the consequence of an event even if the event is exceedingly rare. To begin the analysis, a boolean representation of the event tree is made using the sum of the Minimal Cut Sets followed by the simplification of the equation. The minimum cut sets are the paths that lead to core damage and are taken from Figure 18. The equation begins as follows:

$$
\begin{aligned}
P(CD|I) = {} & P(S)\,P(A)\,P(E)\,\overline{P(R)} \\
& + P(S)\,P(A)\,\overline{P(E)}\,\overline{P(R)} \\
& + P(S)\,\overline{P(A)} \\
& + \overline{P(S)}\,P(A)\,P(L)\,P(E)\,\overline{P(R)} \\
& + \overline{P(S)}\,P(A)\,P(L)\,\overline{P(E)}\,\overline{P(R)} \\
& + \overline{P(S)}\,P(A)\,\overline{P(L)}\,P(E)\,\overline{P(R)} \\
& + \overline{P(S)}\,P(A)\,\overline{P(L)}\,\overline{P(E)} \\
& + \overline{P(S)}\,\overline{P(A)}
\end{aligned} \tag{9}
$$

This equation is then simplified using boolean algebra identifies to get the final CCDP equation below:

$$P(CD|I) = \overline{P(A)} + P(S)P(A)P(L)\overline{P(R)} + \left(\overline{P(E)} + P(E)\overline{P(R)}\right)\left(\overline{P(S)}P(A)P(R)\overline{P(L)}\right) \tag{10}$$
$$= 1.5\text{x}10^{-7} + 0.372 + 2.3\text{x}10^{-4}$$
$$= 0.373$$

5.3 Core Damage Frequency

$$\begin{aligned} F(CD) &= F(I)\ P(CD|I) \\ &= (2,2 \cdot 10^{-5})(0.373) \\ &= 8.2 \cdot 10^{-6} \end{aligned} \tag{11}$$

This number is aligned with the general order of magnitude in plant safety and similar to the one analyzed down by the NRC, in for the Surry NPP [5], the total core damage frequency from all SBO initiating events (SBO+Battery Depletion, SBO+SBLOCA, SBO+AFWfail + SBO+SORV + ...) is $2.7664 \cdot 10^{-5}$ pry. The core damage frequency for the Zion NPP is a similar estimate of $9.34 \cdot 10^{-6}$ from the NRC [6].

6 Safety System Implementation: Passive Secondary Heat Removal System

This section is dedicated to the analysis and implementation of a passive safety system specifically designed to cope with an SBO scenario. A deterministic safety analysis is performerd with the aim to evaluate the performance of the new system into the scenario treated in this report. Although the system was originally designed to activate after the batteries that power the AFW got depleted (i.e after 5 hours in this scenario), two cases are analyzed: with and without the AFW system. The case where the AFW is not available remains the most severe one.

6.1 Description of the system

The Passive Secondary Heat Removal System (PSHRS) has been designed to extend the SBO coping capability of the Zion NPP from 5 h up to the 72 h required by the revised 10 CFR 50.63. The system is fully passive: it does not rely on AC power, AAC sources, active pumps, compressed air, or operator action once the actuation condition is reached.

The PSHRS consists of a closed natural-circulation loop connected in parallel to the Main Steam Line (MSL) and the Main Feedwater Line of each Steam Generator, with all connections placed *outside* the containment in order to avoid additional containment penetrations (GDC 54). The loop routes saturated steam from the SG outlet to a vertical tube-bundle Heat Exchanger (HX) immersed in an Elevated Evaporation Pool (EEP) located above the SG elevation. The steam condenses on the HX tubes, and the condensate returns by gravity to the SG feedwater line, closing the loop. The heat absorbed by the pool is removed by evaporation to the atmosphere through a dedicated vent line at the top of the EEP enclosure. The driving head is provided by the density difference between the saturated vapour in the hot leg and the subcooled condensate in the cold leg, both at the secondary-side pressure of ∼67 bar.

Isolation during normal plant operation is ensured by fail-open electromagnetic valves arranged in a series–parallel configuration that satisfies the single failure criterion: valves required to open

on demand are placed in parallel, while valves required to remain closed during power operation are placed in series. The system is normally kept closed by the DC bus; upon sustained loss of both AC and DC power (filtered to discriminate the short transient at battery switch-over), the valves automatically open and natural circulation establishes itself.
The peak thermal load on the system occurs at t = 5 h after SBO, when the DC batteries are depleted and the AFW Turbine-Driven Pump becomes uncontrolled. This instant defines the worst-case condition for the hydraulic sizing of the loop. The main design parameters are summarised in Table VI.

Table VI: Main design parameters of the PSHRS (per loop).

Parameter	Value	Unit
Reactor thermal power P_0	3250	$\mathrm{MW_{th}}$
Peak decay power at $t = 5\,\mathrm{h}$	31.8	MW
Total decay energy (5–72 h)	4.286×10^{12}	J
Secondary-side pressure	67	bar
Latent heat of vaporisation at 67 bar	1524	kJ/kg
Saturated liquid density at 67 bar	745.28	$\mathrm{kg/m^3}$
Saturated vapour density at 67 bar	34.786	$\mathrm{kg/m^3}$
Required steam mass flow rate per loop	5.22	kg/s
Elevation difference H (SG–HX)	8.5	m
Natural circulation driving pressure	5.92×10^4	Pa
Minimum pipe diameter $D_{\min}$	0.125	m
Steam velocity at $D_{\min}$	12.16	m/s
Reynolds number at $D_{\min}$	2.65×10^6	–
Total pressure losses	5.48×10^4	Pa
Natural circulation margin	1.08	–
Evaporated water mass (5–72 h)	1652.9	t
Required pool volume (incl. 20% margin)	1989.5	$\mathrm{m^3}$

The natural-circulation margin $\Delta p_{\mathrm{drive}}/\Delta p_{\mathrm{loss}} = 1.08$ confirms that the loop is self-driven under worst-case conditions, while the pool inventory is sufficient to absorb the decay energy released between battery depletion and AC restoration at $t = 72\,\mathrm{h}$.
Figure 19 shows the evolution of the decay heat after shutdown. The most of the power is released at the initial hours from the shutdown and therefore are the most severe one.

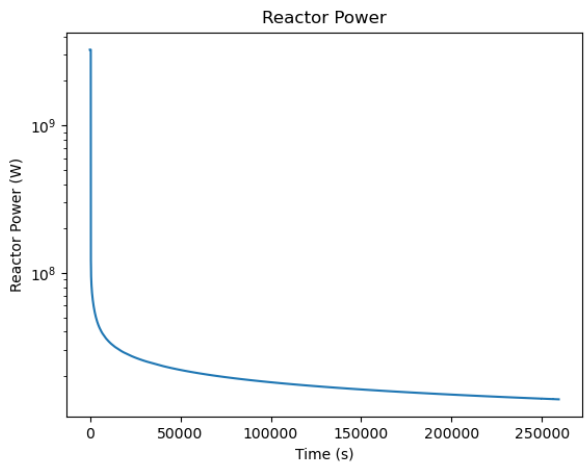

Figure 19: Core Decay Power

6.2 Design Modifications from the Preliminary Configuration

The first RELAP5 simulations of the PSHRS coupled to the Zion model revealed that the system, as initially sized in the hydraulic pre-design report, was not able to prevent core damage within the 72-hour SBO coping period, in neither of the two reference configurations tested (with and without auxiliary feedwater contribution during the first 5 h). Although the hand-calculated natural-circulation margin was positive, the integrated transient behaviour highlighted three limiting factors that had been underestimated by the preliminary lumped-parameter sizing:

- the available heat-exchange surface of the HX bundle was insufficient to condense the steam flow at the rate required by the peak decay heat, leading to a progressive degradation of the secondary heat sink;
- the hydraulic resistance of the inlet pipe to the HX produced a pressure drop comparable to the natural-circulation driving head when evaluated with the realistic two-phase flow distribution computed by RELAP5;
- the EEP water inventory, although sufficient on the basis of the integral decay energy balance, was depleted earlier than predicted because of the higher-than-expected evaporation rate during the initial transient phase, when the primary system is still releasing its stored sensible heat in addition to the decay heat.

In addition, the secondary depressurisation strategy inherited from the EOPs (cooldown down to 15 bar) was found to be detrimental in the long-term passive phase, because the associated steam release through the SG relief valves wasted secondary inventory that would otherwise be available for the natural-circulation loop. As a consequence, several design modifications have been implemented in order to solve this set of problems. The full set of modifications is described in the following paragraphs.

6.2.1 Reduced Secondary Depressurisation Target

The cooldown setpoint of the SG relief valves has been modified from the EOP-prescribed value of 15 bar to a higher target of 53.6 bar. The rationale is twofold: first, a higher secondary pressure preserves the secondary inventory by avoiding the prolonged steam release through the SG relief valves observed in the base-case transient; second, the PSHRS is designed to operate at the secondary saturation pressure of $\sim$67 bar used for the hydraulic sizing, and a moderate cooldown to 53.6 bar − rather than to 15 bar − keeps the loop boundary conditions close to the design point of the heat exchanger, where the condensation performance is maximised. The modified cooldown is still consistent with the safety function of the EOP (bringing the plant to a more benign state), while preventing the unnecessary depletion of the SG water inventory that the long-term passive phase relies upon.

6.2.2 Enlarged HX Inlet Pipe

The cross-sectional flow area of the inlet pipe connecting the Main Steam Line to the HX has been increased from the preliminary value of $0.0122\,\mathrm{m}^2$ to $0.5\,\mathrm{m}^2$ (a factor ~ 40 increase). The preliminary sizing, based on a single-phase turbulent friction estimate at the design steam velocity, severely underestimated the actual two-phase pressure drop computed by RELAP5 along the loop. The enlarged inlet area reduces the steam velocity and the associated frictional losses, restoring a positive natural-circulation margin under the realistic flow distribution and allowing the loop to self-establish without external assistance. This modification is consistent with the IAEA TECDOC-1624 guidance on passive natural-circulation loops, which recommends generous oversizing of the flow areas to compensate for the modelling uncertainties typical of preliminary one-dimensional sizing.

6.2.3 Increased HX Heat-Transfer Area

The total heat-transfer surface of the HX tube bundle has been increased from $0.042\,\mathrm{m}^2$ to $0.61\,\mathrm{m}^2$ per tube (a factor ~ 14.5 increase), achieved by lengthening the tubes and increasing their number. The original sizing was based on a LMTD-type estimate at the peak decay heat condition, but the RELAP5 simulations showed that the actual condensation heat transfer coefficient is significantly lower than the textbook correlation value, especially during the initial transient phase when the pool is still being heated up and the saturation temperature on the pool side is not yet fully developed. The enlarged heat-transfer area provides the additional margin needed to condense the full steam flow at the peak load, and ensures that the temperature difference across the bundle remains within reasonable values throughout the 72-hour transient.

6.2.4 Pool Volume Increased by 50%

The total water inventory of the EEP, in the case without axuliary feedwater, has been increased by a factor 1.5 with respect to the value reported in Table VI, bringing the pool volume from $\sim$2000 m^3 to $\sim$3000 m^3. This additional margin absorbs three effects that were not captured by the preliminary integral energy balance: (i) the higher evaporation rate during the initial passive phase, dominated by the residual sensible heat of the primary system in addition to the decay heat; (ii) the lower-than-ideal HX efficiency, which leads to a higher pool temperature and therefore to a larger fraction of the heat removal being achieved by latent (evaporative) rather than sensible heat-up; (iii) the modelling uncertainties associated with the new flow distribution after the enlargement of the inlet pipe and of the HX area. The increased inventory ensures that the HX bundle remains submerged for the full 72-hour mission time, preserving the heat-removal function until the end of the transient.

6.2.5 T-shaped Evaporation Pool

The EEP has been reshaped from a uniform prismatic geometry to a T-shaped cross-section, with a wider upper section acting as the main evaporation reservoir and a narrower lower section that hosts the Heat Exchanger bundle. This geometry guarantees that the HX remains fully submerged throughout the transient even after a significant fraction of the pool inventory has been boiled off: as the pool level decreases with time, the residual water remains confined in the narrow lower section, preserving the heat transfer surface in contact with liquid water. In the original cylindrical/prismatic design, the same loss of inventory would have produced a faster drop in water level and a premature partial uncovery of the HX tubes, degrading the condensation performance well before the 72-h mission time. The total water inventory is unchanged with respect to the value reported in Table VI, but its vertical distribution is now optimised to maintain HX submergence as the dominant geometric constraint.

6.2.6 Enlarged Atmospheric Vent

The cross-sectional area of the vent line connecting the EEP headspace to the atmosphere has been increased with respect to the preliminary sizing. The objective is to keep the pool effectively at atmospheric pressure during the entire transient, avoiding pressurisation of the EEP enclosure caused by the steam generated by pool boiling. A pressurised pool would raise the saturation temperature of the water, reduce the temperature difference available for condensation on the HX tubes and ultimately reduce the heat removal capability of the system. The enlarged vent ensures that the steam produced at the peak evaporation rate can be discharged without appreciable pressure build-up, keeping the pool boundary condition consistent with the hypothesis used in the sizing calculations ($p_{\mathrm{pool}} \approx p_{\mathrm{atm}}$, $T_{\mathrm{sat}} \approx 100°\mathrm{C}$).

6.2.7 Delayed Actuation (5 h after SBO)

The actuation logic has been tied to the depletion of the DC batteries at $t \approx 5\,\text{h}$ after the initiating event, rather than to the SBO signal itself. During the first 5 h, decay heat removal is already ensured by the existing Turbine-Driven AFW supplied from the Condensate Storage Tank and controlled through DC power. Activating the PSHRS during this phase would be redundant and could interfere with AFW operation, introduce unnecessary pool inventory consumption and complicate the cooldown strategy followed by the operators. The PSHRS is therefore designed to engage exactly when the existing heat-removal chain is about to be lost: the fail-open isolation valves open upon sustained loss of the DC signal, with a time-filter that discriminates the brief AC-to-DC switch-over from a genuine battery depletion. This staged actuation maximises the useful pool inventory available for the long-term passive phase and ensures a smooth hand-over from the active (AFW + batteries) to the passive (PSHRS) heat removal mode.

6.3 Deterministic Safety Analysis: Case without AFW

In accordance with the project statement, the deterministic safety analysis of the PSHRS has been performed on the most demanding sequence among those without RCP seal leakage. The simulated scenario assumes complete unavailability of the AFW system from $t = 0$ (i.e. the PSHRS is the only secondary heat removal path for the entire transient), no AC restoration, and no operator intervention. This represents a conservative envelope: if the PSHRS alone is sufficient to prevent core damage under these conditions, the same conclusion belongs to all the less demanding sequences in which the AFW + TDP contributes for the first 5 h. The simulation has been carried out for the full 72-hour SBO coping period required by 10 CFR 50.63.

6.3.1 Core Level and Peak Cladding Temperature

These parameters are the acceptance criteria that determine whether core damage has occurred or not. Figure 20 shows the collapsed liquid level in the core and in the upper plenum. The core level remains constant at $\sim 4.08\,\text{m}$ for the entire 72-hour transient, well above the top of the active fuel: the core is never uncovered and the cladding integrity is preserved. The upper plenum level shows a transient depression down to $\sim 3.3\,\text{m}$ at $t \approx 23\,\text{h}$, which corresponds to the redistribution of primary inventory during the establishment of the new equilibrium between primary natural circulation and the PSHRS-driven secondary heat removal. After this transient, the upper plenum level recovers and stabilises around 3.6 m, confirming that the primary side reaches a steady two-phase natural-circulation regime sustained by the PSHRS heat sink.

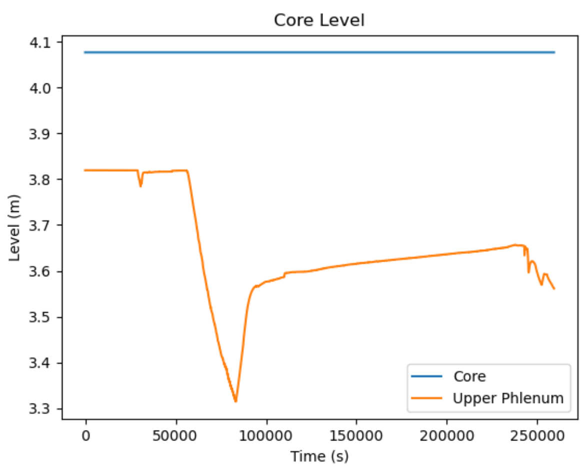

Figure 20: Core Level

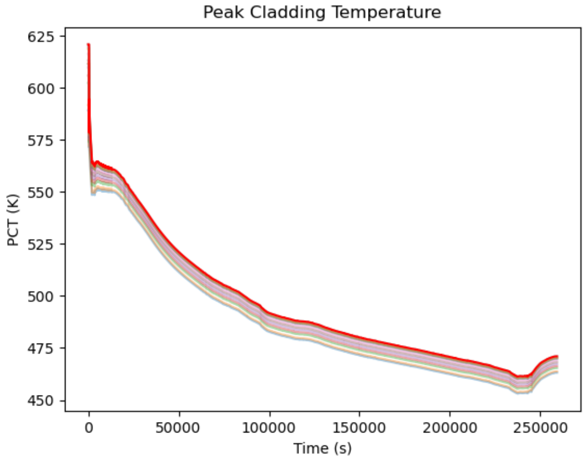

Figure 21: Peak Cladding Temperature Evolution

Figure 21 shows that the acceptance criteria is fully respected as the PCT does not go close to 1477 K, confirming that there is no core damage and the fuel rods are being cooled.

6.3.2 Pool inventory and HX submergence

Figure 22 shows the time evolution of the EEP collapsed liquid level. Starting from an initial value of $\sim 20\,\mathrm{m}$, the pool level decreases monotonically as steam is vented to the atmosphere, reaching $\sim 3.5\,\mathrm{m}$ at $t = 72\,\mathrm{h}$. The initial steeper drop in the first $\sim$8 h corresponds to the higher decay heat load (and hence higher evaporation rate) of the early passive phase, while the slope progressively flattens as the decay power decreases.
The void fraction distribution along the axial nodalisation of the pool, shown in Figure 22, gives a complementary view of this process. The pool has been discretised in 8 axial volumes (1 = top, 8 = bottom). The upper volumes (1, 2) are permanently at high void as they constitute the steam plenum connected to the atmospheric vent. The intermediate volumes void progressively from top to bottom: volume 7 around $t \approx 8\,\mathrm{h}$, volume 6 around $t \approx 28\,\mathrm{h}$, volume 5 around $t \approx 61\,\mathrm{h}$, and volume 4 around $t \approx 67\,\mathrm{h}$. At $t = 72\,\mathrm{h}$, volume 3 still contains liquid water, and crucially the HX bundle remains submerged thanks to the T-shaped pool geometry introduced in Section 6.2: the narrower lower section retains sufficient water inventory to keep the heat exchange surface in contact with liquid even after the bulk of the pool has been boiled off. This confirms the effectiveness of the geometric modification: in a uniform prismatic pool with the same total inventory, the liquid level would have crossed the top of the HX bundle well before 72 h, leading to a premature degradation of the condensation performance.

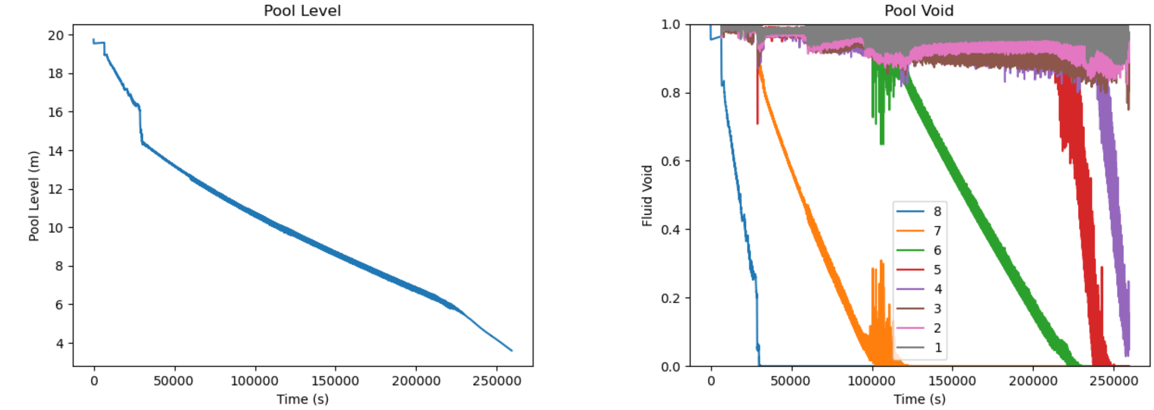

Figure 22: Pool Level (on the left) and Pool Void(on the right).

6.3.3 Absence of AFW contribution and SG Level

Figure 23 confirms that the AFW mass flow is identically zero throughout the simulation. This validates the worst-case assumption adopted in this analysis: the entire decay heat extraction during the 72-hour transient is provided by the PSHRS alone, with no contribution from the active feedwater systems. Due to this reason, the dry-out occurs very early at ≈ 3 hours after the SBO.

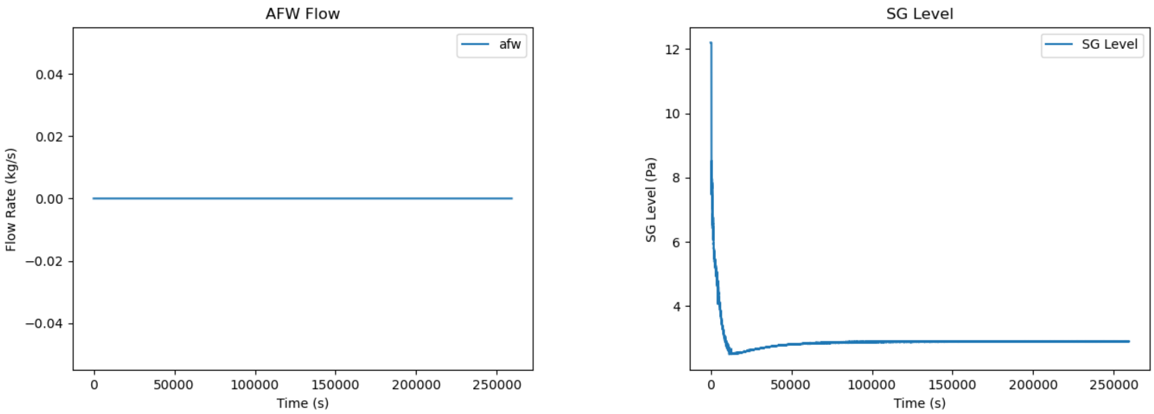

Figure 23: AFW flow (on the left) and SG Level (on the right).

6.3.4 Conclusion of the DSA

The deterministic simulation demonstrates that the PSHRS, acting as the sole secondary heat removal system, is able to:

- maintain core coverage for the full 72-hour SBO coping period, with no approach to the cladding temperature limit;
- establish and sustain a stable two-phase natural circulation in the primary system, supported by the condensation heat sink on the secondary side;
- preserve a sufficient water inventory in the EEP, with the HX bundle still fully submerged at $t = 72\,\text{h}$ thanks to the T-shaped pool design.

The acceptance criteria are therefore satisfied by direct RELAP5 verification. The design is shown to comply with the 72-hour coping requirement of the revised 10 CFR 50.63 even under the conservative assumption of total AFW unavailability from $t = 0$.

6.4 New Core Damage Frequency

The introduction of the PSHRS removes the dependency on AC restoration that dominates the current CCDP. In the modified event tree, all the sequences in which the PSHRS is available become SAFE sequences – as deterministically demonstrated in Section 6.3 for the worst case without RCP seal leakage – regardless of the status of AFW, batteries and AC recovery. The residual CD contribution is therefore associated only with the failure of the PSHRS itself, so that $\mathrm{CCDP_{new}} \approx q_{\mathrm{PSHRS}} \cdot \mathrm{CCDP_{old}}$. Adopting a conservative reliability estimate of $q_{\mathrm{PSHRS}} = 10^{-2}$, consistent with literature values for passive natural-circulation heat removal systems with redundant fail-open isolation valves, the new figures become:

$$\mathrm{CCDP_{new}} \approx 3.73 \times 10^{-3}, \qquad F(CD)_{\mathrm{SBO,new}} \approx 8.21 \times 10^{-8}\ \mathrm{pry}, \tag{12}$$

corresponding to a reduction of approximately two orders of magnitude (i.e. ∼99%) with respect to the current configuration.

6.5 Deterministic Safety Analysis: Realistic Case With AFW Contribution

In addition to the worst-case scenario without AFW analysed in Section 6.3, a second simulation has been carried out under more realistic assumptions: the Turbine-Driven AFW operates during the first 5 h of the transient (until battery depletion), after which the PSHRS takes over as the only available secondary heat removal path. This case corresponds to the actual design intent of the PSHRS and to the most probable accident sequence in which the system is called to operate, and provides a complementary check of the system performance under nominal boundary conditions. We expect to obtain even more optimistic outcomes from this case.

6.5.1 Core Level and Peak Cladding Temperature

Figure 24 shows the collapsed liquid level in the core and in the upper plenum during the 72-hour transient, together with the Peak Cladding Temperature. The core level remains constant at ∼4.08 m for the entire simulation, well above the top of the active fuel: the core is never uncovered, and the cladding integrity is preserved at all times. The upper plenum level, on the other hand, shows a richer dynamic behaviour that reflects the different phases of the transient:

- an initial sharp depression down to ∼2.72 m at $t \approx 6$ h, corresponding to the transition between the active AFW phase and the passive PSHRS phase. When the AFW trips at battery depletion ($t \approx 5$ h), the primary system has to re-establish a new equilibrium between the residual stored heat, the decay power, and the still-developing PSHRS condensation flow. During this short transient, the upper plenum partially voids as steam is generated in the core but the condensation rate in the HX has not yet reached its steady value;

- a stable plateau at ∼3.6 m between $t \approx 14$ h and $t \approx 50$ h, corresponding to the long-term natural-circulation regime sustained by the PSHRS. During this phase the upper plenum operates in a stationary two-phase condition with constant void fraction, which is the design operating point of the passive loop;

- a final progressive drop down to ∼2.5 m between $t \approx 55$ h and $t = 72$ h, associated with the degradation of the HX condensation efficiency as the evaporation pool approaches depletion. As the pool level decreases, the saturation temperature on the pool side rises (the upper volumes of the pool become superheated steam) and the LMTD across the HX decreases, reducing the heat removal rate and increasing the void fraction in the primary upper plenum.

This behaviour is fully consistent with the PCT evolution: the upper plenum drop at $t \approx 6\,\mathrm{h}$ coincides with the PCT drop to $\sim$480 K, the long stable plateau at $\sim$3.6 m coincides with the slow U-shaped PCT recovery between $\sim$15 h and $\sim$55 h, and the final upper-plenum depression mirrors the asymptotic rise of the PCT to $\sim$555 K at the end of the transient. The physical consistency between the two figures confirms the reliability of the RELAP5 results and shows that the PSHRS fullfills the acceptance criteria in this case as it was in the previous one.

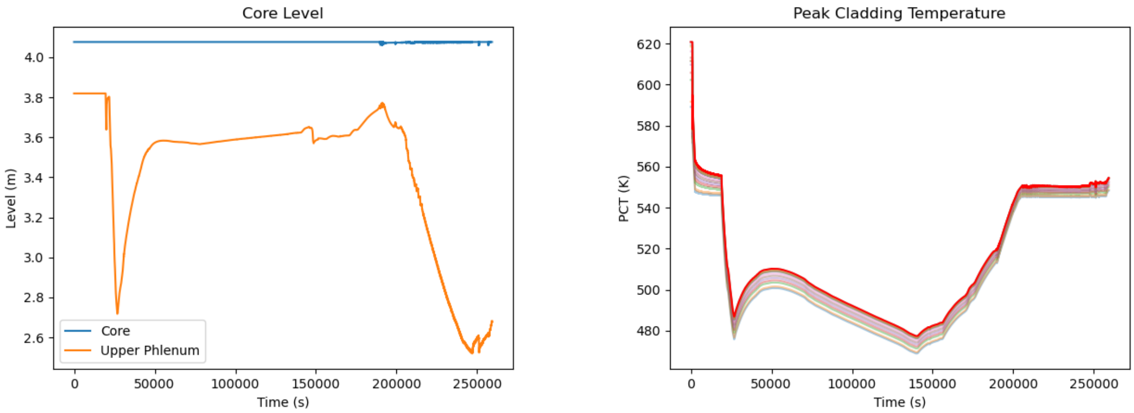

Figure 24: Core Level (on the left) and Peak Cladding Temperature (on the right). Case with AFW.

6.5.2 Pool Inventory: Comparison with the Worst Case

Figure 25 shows the pool collapsed level and the axial void distribution for the realistic case with AFW. The qualitative behaviour is the same as in the worst case without AFW (Section 6.3): the pool drains progressively from the top downwards, with the upper volumes losing their liquid content first as evaporation proceeds and the lower volumes hosting the HX bundle remaining submerged for the longest time, thanks to the T-shaped geometry. The sequential voiding pattern visible in Figure 25 – with the top volumes (1 and 2) reaching high void fraction from the very beginning of the passive phase and the intermediate volumes (3 to 7) draining one after the other over the course of the transient – confirms that the boil-off proceeds in an orderly fashion from top to bottom, as expected for a vented atmospheric pool. The level drops appears faster since the volume of the pool is reduced to the original one in this case, as mentioned in the design modifications chapter.

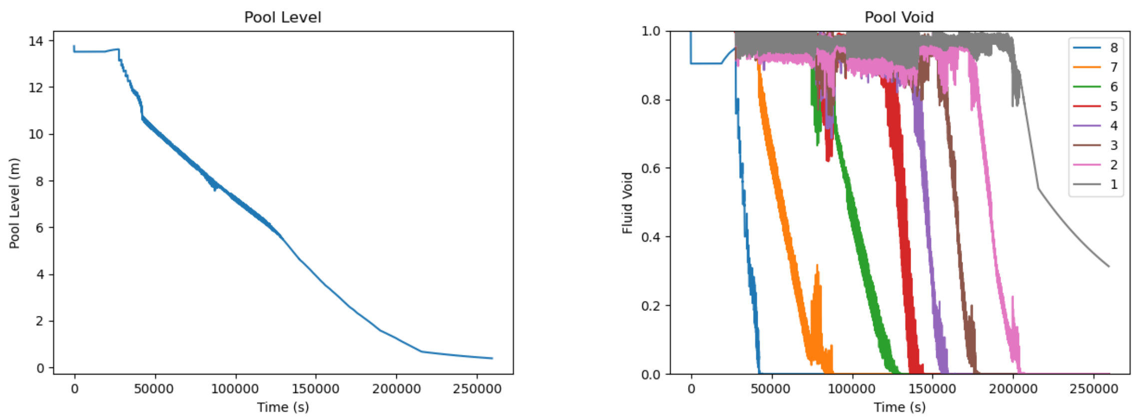

Figure 25: Pool Level (on the left) and Pool Void (on the right). Case with AFW.

6.5.3 SG Inventory and Transition to the PSHRS

Figure 26 shows the collapsed liquid level in the Steam Generator secondary side. The level initially drops to ∼8 m at SCRAM because of the swell collapse, and is then recovered to ∼13.7 m by the AFW during the first 5 h. After the AFW trip, the SG level is maintained at this value by the PSHRS condensate return for ∼50 h, demonstrating that the passive loop is effective at preserving the secondary inventory once established. The level finally starts to decrease around $t \approx 58$ h, when the EEP water inventory is approaching depletion and the condensation rate at the HX begins to degrade. The SG is not yet dry at $t = 72$ h, which means that the secondary heat sink is preserved for the full required duration.

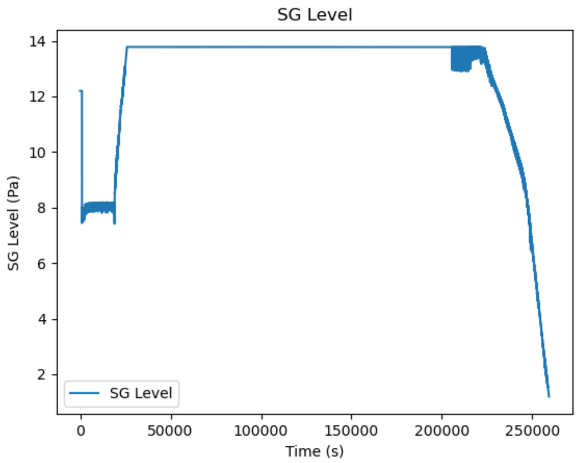

Figure 26: SG Level. Case with AFW.

6.5.4 Conclusion of the realistic case

The simulation with AFW operating during the first 5 h confirms the conclusions drawn from the worst-case analysis: the PSHRS, in its modified configuration, is able to maintain the plant in a safe state for the full 72-hour SBO coping period. The PCT remains ∼925 K below the acceptance limit, the core is never uncovered, the secondary inventory is preserved by the combined effect of AFW and PSHRS condensate return, and the EEP inventory is sufficient to sustain heat removal up to $t = 72$ h even after the conservative pool volume increase by a factor 1.5. The two simulated cases (with and without AFW) bracket the realistic range of operating conditions and demonstrate the robustness of the proposed design with respect to the actual hand-over time between active and passive heat removal.

7 Conclusions

The steady-state calculation is first checked against the target values, and the SBO base case is then used to observe the main thermal-hydraulic response of the plant under blackout conditions.

The SBO base case transient showed that the plant is able to maintain core cooling for at least 12 hours without core damage. The decay heat is effectively removed through natural circulation in the primary system and by the Auxiliary Feedwater system on the secondary side. The peak cladding temperature remains well below the safety limit of 1477 K throughout the entire transient, and the core is never uncovered.
Having analyzed two EOPs, we can say that the SG cooldown procedure (EOP 1) proved usefil but not strictly necessary to prevent core damage within the 12-hour simulation window. The

cooldown EOP reduces primary pressure and lowers the RCP seal leak rate. It is therefore recommended as a prudent operator action. Instead, the primary depressurization procedure (EOP 2) was never triggered in either the base case or the no-cooldown case, as the Core Exit Temperature did not reach the 923 K threshold at any point during the transient. Overall, the results demonstrate that the existing safety systems and operator procedures are sufficient to manage the SBO scenario for 12 hours. However, the analysis also highlights that the absence of the SG cooldown EOP leads to a more challenging scenario and could lead to core damage in a longer event.

The event tree proved that the AC power restoration is the most relevant action in order to avoid core damage in most of the cases analyzed. In every related case, the PCT stays below the safety limit for core damage and the outcome is safe regardless of the seal leak, its amplification or extra-batteries availability. AC restoration is therefore the most important risk reduction factor. Moreover, seal leak amplification has a marginal impact on the system condition. Comparison between cases where the TE3 is present (i.e. ET07 vs ET11, or ET09 vs ET13)shows that the difference in time to core damage is at most 2 hours and the asymptotic PCT is almost identical. This confirm that, in the time scale considered for this scenario, the dominant mechanism of failure is the loss of heat sink rather than the loss of inventory from the RCP seals.

Finally, it has been demonstrated that the PSHRS system is able to cope with these type of scenario, lowering the frequency of core damage. It allows the system to survive to the SBO event in both cases, with and without the AFW, avoiding core damage. Each acceptance criteria has not been overcome (PCT, Core Level) during the transients.

CRediT Author Statement

- **Filippo Dalmonego:** Coding, Writing, Review, Supervision, Visualization.
- **Tyler Ewald:** Coding, Writing, Editing, Supervision.
- **Riccardo Fasano:** Writing, Coding, Editing, Supervision.
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